



4.0.1 UGC NET CSE | June 2012 | Part 3 | Question: 73



In which file the compiler manage various objects, which are used in windows programming?

- A. Control File B. Binary File C. Text File D. Obj File

ugcnetcse-june2012-paper3 compiler-design

Answer key

4.0.2 UGC NET CSE | June 2013 | Part 2 | Question: 21



Given the production rules of a grammar G1 as

$$S_1 \rightarrow AB \mid aaB$$

$$A \rightarrow a \mid Aa$$

$$B \rightarrow b$$

and the production rules of a grammar G2 as

$$S_2 \rightarrow aS_2bS_2 \mid bS_2aS_2 \mid \lambda$$

Which of the following is correct statement?

- A. G1 is ambiguous and G2 is not ambiguous B. G1 is ambiguous and G2 is ambiguous
C. G1 is not ambiguous and G2 is ambiguous D. G1 is not ambiguous and G2 is not ambiguous

ugcnetcse-june2013-paper2 compiler-design

Answer key

4.0.3 UGC NET CSE | June 2013 | Part 2 | Question: 23



The equivalent production rules corresponding to the production rules $S \rightarrow S\alpha_1 \mid S\alpha_2 \mid \beta_1 \mid \beta_2$ is

- A. $S \rightarrow \beta_1 \mid \beta_2, A \rightarrow \alpha_1 A \mid \alpha_2 A \mid \lambda$
B. $S \rightarrow \beta_1 \mid \beta_2 \mid \beta_1 A \mid \beta_2 A, A \rightarrow \alpha_1 A \mid \alpha_2 A$
C. $S \rightarrow \beta_1 \mid \beta_2, A \rightarrow \alpha_1 A \mid \alpha_2 A$
D. $S \rightarrow \beta_1 \mid \beta_2 \mid \beta_1 A \mid \beta_2 A, A \rightarrow \alpha_1 A \mid \alpha_2 A \mid \lambda$

ugcnetcse-june2013-paper2 compiler-design

Answer key

4.0.4 UGC NET CSE | December 2013 | Part 2 | Question: 29



The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned address is called

- A. Symbol resolution B. Parsing C. Assembly D. Relocation

ugcnetcse-dec2013-paper2 compiler-design

Answer key

4.0.5 UGC NET CSE | December 2013 | Part 3 | Question: 17



The compiler converts all operands upto the type of the largest operand is called

- A. Type Promotion B. Type Evaluation
C. Type Conversion D. Type Declaration

ugcnetcse-dec2013-paper3 compiler-design

Answer key

4.0.6 UGC NET CSE | Junet 2015 | Part 2 | Question: 35



Which phase of compiler generates stream of atoms?

- A. Syntax Analysis
- B. Lexical Analysis
- C. Code Generation
- D. Code Optimization

ugcnetcse-june2015-paper2 compiler-design

Answer key

4.0.7 UGC NET CSE | June 2009 | Part 2 | Question: 32



The linker :

- A. is similar to interpreter
- B. uses source code as its input
- C. is required to create a load module
- D. none of the above

ugcnetcse-june2009-paper2

Answer key

4.0.8 Which activity is not included in the first pass of two pass assemblers (UGC NET Dec 04)



Q1. Which activity is not included in the first pass of two pass assemblers?

- (A) Build the symbol table
- (B) Construct the intermediate code
- (C) Separate mnemonic opcode and operand fields
- (D) None of the above

4.0.9 UGC NET CSE | November 2017 | Part 2 | Question: 35



Match the description of several parts of a classic optimizing compiler in **List-I**, with the names of those parts in **List-II**:

List-I

- (a) A part of a compiler that is responsible for recognizing syntax.
- (b) A part of a compiler that takes as input a stream of characters and produces as output a stream of words along with their associated syntactic categories.
- (c) A part of a compiler that understand the meanings of variable names and other symbols and checks that they are used in ways consistent with their definitions.
- (d) An IR-to-IR transformer that tries to improve the IR program in some way (Intermediate Representation).

List-II

- (i) Optimizer
- (ii) Semantic Analysis
- (iii) Parser
- (iv) Scanner

Code :

- A. (a)-(iii); (b)-(iv); (c)-(ii); (d)-(i)
- B. (a)-(iv); (b)-(iii); (c)-(ii); (d)-(i)
- C. (a)-(ii); (b)-(iv); (c)-(i); (d)-(iii)
- D. (a)-(ii); (b)-(iv); (c)-(iii); (d)-(i)

Answer key **4.0.10 UGC NET CSE | June 2006 | Part 2 | Question: 4**

The logic of pumping lemma is a good example of :

- | | |
|---------------------------------|--------------|
| A. pigeon hole principle | B. recursion |
| C. divide and conquer technique | D. iteration |

ugcnetcse-june2006-paper2 compiler-design

Answer key **4.0.11 UGC NET CSE | December 2004 | Part 2 | Question: 31**

Which activity is not included in the first pass of two pass assemblers ?

- | | |
|--|------------------------------------|
| A. Build the symbol table | B. Construct the intermediate code |
| C. Separate mnemonic opcode and operand fields | D. None of the above |

ugcnetcse-dec2004-paper2

Answer key **4.0.12 UGC NET CSE | December 2004 | Part 2 | Question: 35**

In two pass assembler the symbol table is used to store :

- | | | | |
|--------------------|---------------|-------------|--------------------|
| A. Label and value | B. Only value | C. Mnemonic | D. Memory Location |
|--------------------|---------------|-------------|--------------------|

ugcnetcse-dec2004-paper2

Answer key **4.0.13 UGC NET CSE | December 2004 | Part 2 | Question: 34**

Which activity is included in the first pass of two pass assemblers?

- | | |
|--|------------------------------------|
| A. Build the symbol table | B. Construct the intermediate code |
| C. Separate mnemonic opcode and operand fields | D. None of these |

ugcnetcse-dec2004-paper2

Answer key **4.0.14 UGC NET CSE | December 2004 | Part 2 | Question: 33**

Code optimization is responsibility of :

- | | |
|---------------------------|----------------------|
| A. Application programmer | B. System programmer |
| C. Operating system | D. All of the above |

ugcnetcse-dec2004-paper2

Answer key **4.0.15 UGC NET CSE | December 2006 | Part 2 | Question: 39**

A permanent database of a general model of compiler is _____ .

- | | |
|---------------------|-------------------|
| A. Identifier table | B. Page map table |
| C. Literal table | D. Terminal table |

ugcnetcse-dec2006-paper2

4.0.16 UGC NET CSE | December 2006 | Part 2 | Question: 33

A compiler for a high level language that runs on one machine and produces code for a different machine is called :

- A. Optimizing
- C. Cross compiler

- B. One pass compiler
- D. Multipass compiler

ugcnetcse-dec2006-paper2

Answer key 

4.0.17 UGC NET CSE | December 2006 | Part 2 | Question: 32



Which of the following is the most general phase-structured grammar ?

- A. Regular
- B. Context-sensitive
- C. Context free
- D. Syntax tree

ugcnetcse-dec2006-paper2

Answer key 

4.0.18 UGC NET CSE | December 2007 | Part 2 | Question: 35



In the context of compiler design, “reduction in strength” refers to :

- A. code optimization obtained by the use of cheaper machine instructions
- B. reduction in accuracy of the output
- C. reduction in the range of values of input variables
- D. reduction in efficiency of the program

ugcnetcse-dec2007-paper2

Answer key 

4.0.19 UGC NET CSE | December 2007 | Part 2 | Question: 34



Top-down parsers are predictive parsers, because :

- A. next tokens are predicted
- B. length of the parse tree is predicted before hand
- C. lowest node in the parse tree is predicted
- D. next lower level of the parse tree is predicted

ugcnetcse-dec2007-paper2

Answer key 

4.0.20 UGC NET CSE | December 2007 | Part 2 | Question: 33



Absolute loader demands that the programmer needs to know the :

- A. start address of the available main memory
- B. total size of the program
- C. actual address of the data location
- D. absolute values of the operands used

ugcnetcse-dec2007-paper2

Answer key 

4.0.21 UGC NET CSE | December 2007 | Part 2 | Question: 32



A single instruction in an assembly language program contains :

- A. one micro operation
- B. one macro operation
- C. one instruction to be completed in a single pulse
- D. one machine code instruction

ugcnetcse-dec2007-paper2

Answer key 

4.0.22 UGC NET CSE | December 2007 | Part 2 | Question: 31



In a two pass compiler, during the first pass :

- A. user defined address symbols are correlated with their binary equivalent
- B. the syntax of the statement is checked and mistakes, if any, are listed
- C. object program is generated
- D. semantic of the source program is elucidated

ugcnetcse-dec2007-paper2

Answer key

4.0.23 UGC NET CSE | October 2020 | Part 2 | Question: 74



Find the lexicographic ordering of the bit strings given below based on the ordering $0 < 1$.

- a. 001
- b. 010
- c. 011
- d. 0001
- e. 0101

Choose the correct answer from the options given below:

- A. $001 < 010 < 011 < 0001 < 0101$
- B. $0001 < 001 < 010 < 0101 < 011$
- C. $0001 < 0101 < 001 < 010 < 011$
- D. $001 < 010 < 0001 < 0101 < 011$

ugcnetcse-oct2020-paper2 compiler-design

Answer key

4.1

Assembler (2)

4.1.1 Assembler: UGC NET CSE | December 2005 | Part 2 | Question: 35



Which activities is not included in the first pass of two pass assembler?

- A. build the symbol table
- B. construct the Intermediate code
- C. separate mnemonic opcode and operand field.
- D. none of these

ugcnetcse-dec2005-paper2 compiler-design assembler

Answer key

4.1.2 Assembler: UGC NET CSE | June 2005 | Part 2 | Question: 33



Which activity is not included in the first pass of two pass assemblers?

- A. Build the symbol table
- B. Construct the machine code
- C. Separate mnemonic opcode and operand fields.
- D. None of these

ugcnetcse-june2005-paper2 compiler-design assembler

Answer key

4.2

Chomsky Normal Form (1)

4.2.1 Chomsky Normal Form: UGC NET CSE | December 2012 | Part 3 | Question: 53



If the parse tree of a word w generated by a Chomsky form grammar has no path of length greater than i , then the word w is of length

- A. no greater than 2^{i+1}
C. no greater than 2^{i-1}

- B. no greater than 2^i
D. no greater than i

chomsky-normal-form ugcnetcse-dec2012-paper3 parsing

Answer key 

4.3

Code Optimization (2)

4.3.1 Code Optimization: UGC NET CSE | December 2006 | Part 2 | Question: 35



Peer-hole optimization is a form of :

- A. loop optimization
C. constant folding
B. local optimization
D. data flow analysis

ugcnetcse-dec2006-paper2 compiler-design code-optimization

Answer key 

4.3.2 Code Optimization: UGC NET CSE | December 2012 | Part 2 | Question: 11



In compiler design 'reducing the strength' refers to

- A. reducing the range of values of input variables
C. reducing efficiency of program
B. code optimization using cheaper machine instructions
D. none of the above

code-optimization ugcnetcse-dec2012-paper2 compiler-design

Answer key 

4.4

Compilation Phases (2)

4.4.1 Compilation Phases: UGC NET CSE | August 2016 | Part 2 | Question: 33



In compiler optimization, operator strength reduction uses mathematical identities to replace slow math operations with faster operations. Which of the following code replacements is an illustration of operator strength reduction ?

- A. Replace $P + P$ by $2 * P$ or Replace $3 + 4$ by 7.
B. Replace $P * 32$ by $P << 5$
C. Replace $P * 0$ by 0
D. Replace $(P << 4) - P$ by $P * 15$

ugcnetcse-aug2016-paper2 compiler-design compilation-phases

Answer key 

4.4.2 Compilation Phases: UGC NET CSE | June 2005 | Part 2 | Question: 31



Which of the statements related to Compilers is wrong ?

- A. Lexical analysis is breaking the input into tokens
C. Syntax analysis is for analyzing the semantic
B. Syntax analysis is for parsing the phrase
D. None of these

ugcnetcse-june2005-paper2 compiler-design compilation-phases

Answer key 

4.5

Compiler Optimization (2)

4.5.1 Compiler Optimization: UGC NET CSE | December 2015 | Part 2 | Question: 41



Loop unrolling is a code optimization technique:

- A. that avoids tests at every iteration of the loop
B. that improves performance by decreasing the number of instructions in a basic block
C. that exchanges inner loops with outer loops
D. that reorders operations to allow multiple computations to happen in parallel

Answer key **4.5.2 Compiler Optimization: UGC NET CSE | July 2016 | Part 2 | Question: 33**

In ____, the bodies of the two loops are merged together to form a single loop provided that they do not make any references to each other.

- A. Loop unrolling
C. Loop concatenation
- B. Strength reductions
D. Loop jamming

ugcnetcse-july2016-paper2 compiler-design compiler-optimization

Answer key **4.6 Constant Folding (1)****4.6.1 Constant Folding: UGC NET CSE | June 2019 | Part 2 | Question: 73**

Replacing the expression $4 * 2 \cdot 14$ by $8 \cdot 56$ is known as

- A. constant folding
C. strength reduction
- B. induction variable
D. code reduction

ugcnetcse-june2019-paper2 code-optimization constant-folding

Answer key **4.7 Context Sensitive (1)****4.7.1 Context Sensitive: UGC NET CSE | June 2005 | Part 2 | Question: 32**

Which of the following is the most general phase – structured grammar ?

- A. Regular
C. Context free
- B. Context – Sensitive
D. None of these

ugcnetcse-june2005-paper2 compiler-design grammar context-sensitive

Answer key **4.8 Control Flow Graph (1)****4.8.1 Control Flow Graph: UGC NET CSE | December 2013 | Part 3 | Question: 4**

Given a flow graph with 10 nodes, 13 edges and one connected components, the number of regions and the number of predicate (decision) nodes in the flow graph will be

- A. 4, 5
B. 5, 4
C. 3, 1
D. 13, 8

ugcnetcse-dec2013-paper3 compiler-design control-flow-graph

Answer key **4.9 Grammar (6)****4.9.1 Grammar: UGC NET CSE | December 2009 | Part 2 | Question: 32**

Which of the following grammar is LR (1) ?

- (A) $A \rightarrow a A b, A \rightarrow b A b, A \rightarrow a, A \rightarrow b$
- (B) $A \rightarrow a A a, A \rightarrow a A b, A \rightarrow c$
- (C) $A \rightarrow A + A, A \rightarrow a$
- (D) Both (A) and (B)

4.9.2 Grammar: UGC NET CSE | December 2009 | Part 2 | Question: 35

Synthesized attribute can be easily simulated by a

- (A) LL grammar
- (B) Ambiguous grammar
- (C) LR grammar
- (D) None of the above

ugcnetcse-dec2009-paper2 compiler-design grammar

Answer key

4.9.3 Grammar: UGC NET CSE | December 2015 | Part 3 | Question: 24

The language of all non-null strings of a's can be defined by a context free grammar as follow :

$$S \rightarrow aS \mid Sa \mid a$$

The word a^3 can be generated by _____ different trees.

- A. Two
- B. Three
- C. Four
- D. Five

compiler-design grammar ugcnetcse-dec2015-paper3

Answer key

4.9.4 Grammar: UGC NET CSE | June 2010 | Part 2 | Question: 34

Which of the following is the most general phase structured grammar ?

- A. Regular
- B. Context-sensitive
- C. Context free
- D. None of the above

ugcnetcse-june2010-paper2 compiler-design grammar

Answer key

4.9.5 Grammar: UGC NET CSE | June 2014 | Part 2 | Question: 06

A grammar G is $LL(1)$ if and only if the following conditions hold for two distinct productions $A \rightarrow \alpha \mid \beta$

I. $\text{First}(\alpha) \cap \text{First}(\beta) \neq \{a\}$ where a is some terminal symbol of the grammar.

II. $\text{First}(\alpha) \cap \text{First}(\beta) \neq \lambda$

III. $\text{First}(\alpha) \cap \text{Follow}(A) = \phi$ if $\lambda \in \text{First}(\beta)$

- A. I and II
- B. I and III
- C. II and III
- D. I, II and III

ugcnetcse-june2014-paper2 compiler-design grammar parsing

Answer key

4.9.6 Grammar: UGC NET CSE | October 2022 | Part 1 | Question: 83

Consider the grammar $S \rightarrow SbS \mid a$.

Consider the following statements:

The string abababa has

- (A) two parse trees
- (B) two left most derivations
- (C) two right most derivations

Which of the following is correct?

- A. All (A), (B) and [(C)] are true,
- B. Only (B) is true

C. Only (C) is true

D. Only (A) is true

ugcnetcse-oct2022-paper1 compiler-design grammar easy

4.10

Intermediate Code (1)

4.10.1 Intermediate Code: UGC NET CSE | October 2020 | Part 2 | Question: 30



Which of the following is not an intermediate code form?

- A. Syntax trees
- C. Quadruples

- B. Three address codes
- D. Post fix Notation

ugcnetcse-oct2020-paper2 compiler-design intermediate-code

Answer key

4.11

LR Parser (2)

4.11.1 LR Parser: UGC NET CSE | December 2006 | Part 2 | Question: 34



The 'K' in LR (K) cannot be :

A. 0

B. 1

C. 2

D. None of these

ugcnetcse-dec2006-paper2 lr-parser compiler-design

Answer key

4.11.2 LR Parser: UGC NET CSE | December 2014 | Part 2 | Question: 32



Which of the following is true ?

- A. Canonical LR parser is $LR(1)$ parser with single look ahead terminal
- B. All $LR(K)$ parsers with $K > 1$ can be transformed into $LR(1)$ parsers.
- C. Both (A) and (B)
- D. None of the above

ugcnetcse-dec2014-paper2 compiler-design lr-parser

Answer key

4.12

Lexical Analysis (3)

4.12.1 Lexical Analysis: UGC NET CSE | December 2014 | Part 3 | Question: 20



How many tokens will be generated by the scanner for the following statement ?

$x = x * (a + b) - 5;$

A. 12

B. 11

C. 10

D. 07

ugcnetcse-dec2014-paper3 compiler-design lexical-analysis

Answer key

4.12.2 Lexical Analysis: UGC NET CSE | January 2017 | Part 2 | Question: 33



Consider the following statements related to compiler construction:

- I. Lexical Analysis is specified by context-free grammars and implemented by pushdown automata.
- II. Syntax Analysis is specified by regular expressions and implemented by finite-state machine.

Which of the above statement(s) is/are correct?

A. Only I

B. Only II

C. Both I and II

D. Neither I nor II

ugcnetjan2017ii compiler-design lexical-analysis

Answer key 

4.12.3 Lexical Analysis: UGC NET CSE | June 2010 | Part 2 | Question: 35



Which of the following is used for grouping of characters into tokens (in a computer) ?

- A. A parser B. Code optimizer C. Code generator D. Scanner

ugcnetcse-june2010-paper2 compiler-design lexical-analysis

Answer key 

4.13

Linker (1)

4.13.1 Linker: UGC NET CSE | August 2016 | Part 2 | Question: 34



Which of the following are the principles tasks of the linker ?

- I. Resolve external references among separately compiled program units.
- II. Translate assembly language to machine code.
- III. Relocate code and data relative to the beginning of the program.
- IV. Enforce access-control restrictions on system libraries.

- A. *I* and *II* B. *I* and *III* C. *II* and *III* D. *I* and *IV*

ugcnetcse-aug2016-paper2 compiler-design linker

Answer key 

4.14

Machine Instruction (1)

4.14.1 Machine Instruction: UGC NET CSE | December 2015 | Part 2 | Question: 42



What will be the hexadecimal value in the register *ax* (32-bit) after executing the following instructions?

```
mov al, 15
mov ah, 15
xor al, al
mov cl, 3
shr ax, cl
```

- A. 0F00 h B. 0F0F h C. 01E0 h D. FFFF h

ugcnetcse-dec2015-paper2 machine-instruction compiler-design

Answer key 

4.15

Parameter Passing (2)

4.15.1 Parameter Passing: UGC NET CSE | August 2016 | Part 3 | Question: 21



Assume that the program '*P*' is implementing parameter passing with 'call by reference'. What will be printed by following print statements in *P*?

```
Program P
{
    x=10;
    y=3;
    funb (y, x, x)
    print x;
    print y;
}
funb (x, y, z)
{
    y=y+4;
    z=x+y+z;
}
```

A. 10, 7

B. 31, 3

C. 10, 3

D. 31, 7

ugcnetcse-aug2016-paper3 parameter-passing

Answer key 

4.15.2 Parameter Passing: UGC NET CSE | December 2014 | Part 3 | Question: 19



Consider the following statements S_1, S_2 and S_3 :

S_1 : In call-by-value, anything that is passed into a function call is unchanged in the caller's scope when the function returns.

S_2 : In call-by-reference, a function receives implicit reference to a variable used as argument.

S_3 : In call-by-reference, caller is unable to see the modified variable used as argument.

A. S_3 and S_2 are true.

B. S_3 and S_1 are true.

C. S_2 and S_1 are true.

D. S_1, S_2, S_3 are true.

ugcnetcse-dec2014-paper3 compiler-design runtime-environment parameter-passing

Answer key 

4.16

Parsing (13)

4.16.1 Parsing: UGC NET CSE | December 2005 | Part 2 | Question: 32



A Top - down Parse generates

A. Left most derivation

B. Right – most derivation

C. Right most derivation in reverse

D. Left – most derivation in reverse

ugcnetcse-dec2005-paper2 compiler-design parsing

Answer key 

4.16.2 Parsing: UGC NET CSE | December 2009 | Part 2 | Question: 33



A shift-reduce parser carries out the actions specified within braces immediately after reducing with the corresponding rule of the grammar.

$S \rightarrow x x W [\text{print "1"}]$

$S \rightarrow y [\text{print "2"}]$

$W \rightarrow S2 [\text{print "3"}],$

what is the translation of " $x x x y z z$ " ?

(A) 1 1 2 3 1

(B) 1 1 2 3 3

(C) 2 3 1 3 1

(D) 2 3 3 2 1

ugcnetcse-dec2009-paper2 compiler-design parsing

4.16.3 Parsing: UGC NET CSE | December 2012 | Part 2 | Question: 18



Given the following expressions of a grammar

$E \rightarrow E * F / F + E / F$

$F \rightarrow F - F / id$

Which of the following is true?

A. $*$ has the higher precedence than $+$

B. $-$ has the higher precedence than $*$

C. $+$ and $-$ have same precedence

D. $+$ has the higher precedence than $*$

ugcnetcse-dec2012-paper2 compiler-design parsing

Answer key 

4.16.4 Parsing: UGC NET CSE | December 2012 | Part 2 | Question: 27



Which of the following is true while converting CFG to an LL(1) grammar?

- A. Removing left recursion alone
- B. Factoring the grammar alone
- C. Removing left recursion and factoring the grammar
- D. None of the above

ugcnetcse-dec2012-paper2 compiler-design parsing

Answer key 

4.16.5 Parsing: UGC NET CSE | December 2013 | Part 2 | Question: 26



Given the following statements:

S_1 : SLR uses follow information to guide reductions. In case of LR and LALR parsers, the look-aheads are associated with the items and they make use of the left context available to the parser.

S_2 : LR grammar is a large subclass of context free grammar as compared to that SLR and LALR grammars.

Which of the following is true?

- A. S_1 is not correct and S_2 is not correct.
- B. S_1 is not correct and S_2 is correct.
- C. S_1 is correct and S_2 is not correct.
- D. S_1 is correct and S_2 is correct

ugcnetcse-dec2013-paper2 compiler-design parsing

Answer key 

4.16.6 Parsing: UGC NET CSE | December 2015 | Part 2 | Question: 43



Which of the following statements is false?

- A. Top-down parsers are LL parsers where first L stands for left-to-right scan and second L stands for a leftmost derivation
- B. $(000)^*$ is a regular expression that matches only strings containing an odd number of zeroes, including the empty string
- C. Bottom-up parsers are in the LR family, where L stands for left-to-right scan and R stands for rightmost derivation
- D. The class of context-free languages is closed under reversal. That is, if L is any context-free language, then the language $L^R = \{w^R : w \in L\}$ is context-free

ugcnetcse-dec2015-paper2 compiler-design parsing

Answer key 

4.16.7 Parsing: UGC NET CSE | December 2018 | Part 2 | Question: 39



Consider the following grammar G :

- $S \rightarrow A \mid B$
- $A \rightarrow a \mid c$
- $B \rightarrow b \mid c$

where $\{S, A, B\}$ is the set of non-terminals, $\{a, b, c\}$ is the set of terminals.

Which of the following statement(s) is/are correct?

- S_1 : $LR(1)$ can parse all strings that are generated using grammar G .
- S_2 : $LL(1)$ can parse all strings that are generated using grammar G .

Choose the correct answer from the code given below:

- A. Only S_1
- B. Only S_2
- C. Both S_1 and S_2
- D. Neither S_1 nor S_2

Answer key **4.16.8 Parsing: UGC NET CSE | July 2016 | Part 2 | Question: 35**

Which of the following is false?

- A. The grammar $S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$, where S is the only non-terminal symbol and ϵ is the null string, is ambiguous
- B. SLR is powerful than LALR
- C. An LL(1) parser is a top-down parser
- D. YACC tool is an LALR(1) parser generator

Answer key **4.16.9 Parsing: UGC NET CSE | July 2018 | Part 2 | Question: 39**

A bottom-up parser generates:

- A. Left-most derivation in reverse
- B. Right-most derivation in reverse
- C. Left-most derivation
- D. Right-most derivation

Answer key **4.16.10 Parsing: UGC NET CSE | June 2014 | Part 2 | Question: 07**Which of the following suffices to convert an arbitrary CFG to an $LL(1)$ grammar ?

- A. Removing left recursion alone
- B. Removing the grammar alone
- C. Removing left recursion and factoring the grammar
- D. None of the above

Answer key **4.16.11 Parsing: UGC NET CSE | June 2014 | Part 2 | Question: 08**

A shift reduce parser suffers from

- A. Shift reduce conflict only
- B. Reduce reduce conflict only
- C. Both shift reduce conflict and reduce reduce conflict
- D. Shift handle and reduce handle conflicts

Answer key **4.16.12 Parsing: UGC NET CSE | June 2023 | Part 2: 27**

Which of the following parser is most powerful parser?

- A. Operator precedence
- B. SLR
- C. Canonical LR
- D. LALR

Answer key **4.16.13 Parsing: UGC NET CSE | Junet 2015 | Part 2 | Question: 34**

Which one of the following is false?

- A. LALR parser is Bottom-Up parser

- B. A parsing algorithm which performs a left to right scanning and a right most deviation is RL(1)
- C. LR parser is Bottom-Up parser
- D. In LL(1), the 1 indicates that there is a one symbol look-ahead

ugcnetcse-june2015-paper2 compiler-design parsing

Answer key 

4.17 Quadruple Representation (1)

4.17.1 Quadruple Representation: UGC NET CSE | June 2019 | Part 2 | Question: 72



On translating the expression given below into quadruple representation, how many operations are required?

$$(i * j) + (e + f) * (a * b + c)$$

- A. 5
- B. 6
- C. 3
- D. 7

ugcnetcse-june2019-paper2 intermediate-code quadruple-representation

Answer key 

4.18 Runtime Environment (1)

4.18.1 Runtime Environment: UGC NET CSE | January 2017 | Part 2 | Question: 35



Which of the following statement(s) regarding a linker software is/are true ?

- I. A function of a linker is to combine several object modules into a single load module.
 - II. A function of a linker is to replace absolute references in an object module by symbolic references to locations in other modules.
- A. Only I
 - B. Only II
 - C. Both I and II
 - D. Neither I nor II

ugcnetjan2017ii runtime-environment compiler-design

Answer key 

4.19 Shift Reduce Parser (2)

4.19.1 Shift Reduce Parser: UGC NET CSE | December 2014 | Part 2 | Question: 31



Shift-Reduce parsers perform the following :

- A. Shift step that advances in the input stream by $K(K > 1)$ symbols and Reduce step that applies a completed grammar rule to some recent parse trees, joining them together as one tree with a new root symbol.
- B. Shift step that advances in the input stream by one symbol and Reduce step that applies a completed grammar rule to some recent parse trees, joining them together as one tree with a new root symbol.
- C. Shift step that advances in the input stream by $K(K = 2)$ symbols and Reduce step that applies a completed grammar rule to form a single tree.
- D. Shift step that does not advance in the input stream and Reduce step that applies a completed grammar rule to form a single tree.

ugcnetcse-dec2014-paper2 compiler-design shift-reduce-parser

Answer key 

4.19.2 Shift Reduce Parser: UGC NET CSE | June 2019 | Part 2 | Question: 74



Shift-reduce parser consists of

- i. input buffer

- ii. stack
- iii. parse table

Choose the correct option from those given below:

- A. a and b only B. a and c only C. c only D. a, b and c

ugcnetcse-june2019-paper2 shift-reduce-parser

Answer key 

4.20 Symbol Table (3)

4.20.1 Symbol Table: UGC NET CSE | June 2005 | Part 2 | Question: 35



Symbol Table can be used for :

- A. Checking type compatibility B. Suppressing duplication of error message
C. Storage allocation D. All of these

ugcnetcse-june2005-paper2 compiler-design symbol-table

Answer key 

4.20.2 Symbol Table: UGC NET CSE | June 2019 | Part 2 | Question: 71



Which data structure is used by the compiler for managing variables and their attributes?

- A. Binary tree B. Link list C. Symbol table D. Parse table

ugcnetcse-june2019-paper2 compiler-design symbol-table

Answer key 

4.20.3 Symbol Table: UGC NET CSE | October 2020 | Part 2 | Question: 41



Which of the following are applications of symbol table?

- i. Storage allocation
- ii. Checking type compatibility
- iii. Suppressing duplicate error messages

Choose the correct answer from the options given below:

- A. (i) and (ii) only B. (i) and (iii) only
C. (ii) and (iii) only D. (i) (ii) and (iii)

ugcnetcse-oct2020-paper2 compiler-design symbol-table

Answer key 

4.21 System Software& Compilers (1)

4.21.1 System Software& Compilers: UGC NET CSE | June 2010 | Part 2 | Question: 31



In an absolute loading scheme, which loader function is accomplished by a loader ?

- A. Re-allocation B. Allocation C. Linking D. Loading

ugcnetcse-june2010-paper2 system-software&-compilers compiler-design runtime-environment

Answer key 

4.22 Translator (1)

4.22.1 Translator: UGC NET CSE | Junet 2015 | Part 2 | Question: 32

The translator which performs macro calls expansion is called

- A. Macro processor
 C. Macro pre-processor
 B. Micro pre-processor
 D. Dynamic linker

ugcnetcse-june2015-paper2 translator preprocessor

Answer key

4.23**Two Pass Assembler (1)****4.23.1 Two Pass Assembler: UGC NET CSE | December 2014 | Part 2 | Question: 33**

In a two-pass assembler, symbol table is

- A. Generated in first pass
 C. Not generated at all
 B. Generated in second pass
 D. Generated and used only in second pass

ugcnetcse-dec2014-paper2 two-pass-assembler

Answer key

Answer Keys

4.0.1	D	4.0.2	B	4.0.3	D	4.0.4	D	4.0.5	A
4.0.6	B	4.0.7	Q-Q	4.0.8	Q-Q	4.0.9	A	4.0.10	Q-Q
4.0.11	Q-Q	4.0.12	Q-Q	4.0.13	Q-Q	4.0.14	Q-Q	4.0.15	Q-Q
4.0.16	C	4.0.17	Q-Q	4.0.18	Q-Q	4.0.19	Q-Q	4.0.20	Q-Q
4.0.21	Q-Q	4.0.22	Q-Q	4.0.23	B	4.1.1	Q-Q	4.1.2	Q-Q
4.2.1	C	4.3.1	Q-Q	4.3.2	B	4.4.1	Q-Q	4.4.2	Q-Q
4.5.1	A	4.5.2	D	4.6.1	A	4.7.1	Q-Q	4.8.1	B
4.9.1	Q-Q	4.9.2	Q-Q	4.9.3	D	4.9.4	Q-Q	4.9.5	D
4.9.6	Q-Q	4.10.1	X	4.11.1	Q-Q	4.11.2	C	4.12.1	A
4.12.2	D	4.12.3	Q-Q	4.13.1	Q-Q	4.14.1	C	4.15.1	Q-Q
4.15.2	C	4.16.1	A	4.16.2	Q-Q	4.16.3	B	4.16.4	C
4.16.5	D	4.16.6	B	4.16.7	D	4.16.8	B	4.16.9	Q-Q
4.16.10	D	4.16.11	C	4.16.12	Q-Q	4.16.13	B	4.17.1	B
4.18.1	A	4.19.1	B	4.19.2	D	4.20.1	Q-Q	4.20.2	C
4.20.3	D	4.21.1	Q-Q	4.22.1	C	4.23.1	A		